

EXPERIMENT 3

AIM: Write a Program to implement data link layer farming method checksum.

SOLUTION: This program implements the **Checksum Error Detection** technique used in the **Data Link Layer**.

The checksum method is used to detect errors that may occur during data transmission. The sender divides the data into equal-sized words (8-bit values in this program), adds all the data words, and if a carry is generated, the carry is added back to the sum (end-around carry).

Then, the **1's complement** of the final sum is calculated. This value is called the **checksum**.

The checksum is attached to the data and transmitted to the receiver.

At the receiver side, the received data words and the checksum are added together. If a carry is generated, it is again added back to the sum. The receiver then finds the **1's complement** of the final result. If the result is **00000000 (0)**, it indicates that **no error** has occurred during transmission, and the **data is accepted**. Otherwise, an **error is detected**, and the **data is rejected**.

This program demonstrates both the sender-side checksum generation and the receiver-side checksum verification using 8-bit checksum with end-around carry.

Source Code:

```
#include <stdio.h>
#include <conio.h>

void main()
{
    int data[20], rdata[20];
    int n, i;
    unsigned int sum = 0, receivedSum = 0;
    unsigned char checksum;
    int receivedChecksum;

    clrscr();

    /* ----- Sender Side ----- */

    printf("----- Sender Side ----- \n");

    printf("Enter the number of data words: ");
    scanf("%d", &n);

    printf("Enter the data words (8 bit: 0 to 255): \n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &data[i]);

        if(data[i] < 0 || data[i] > 255)
        {
            printf("Invalid input! Enter values between 0 and 255. \n");
            return;
        }
        sum = sum + data[i];

        /* Add carry if generated */
```

```

        if(sum > 255)
        {
            sum = (sum & 255) + 1;
        }
    }

    /* Find 1's complement */
    checksum = (unsigned char)(~sum);

    printf("\nSum = %u", sum);
    printf("\nChecksum = %u", checksum);

    printf("\n\nData and Checksum are transmitted.\n");

    /* ----- Receiver Side ----- */

    printf("\n----- Receiver Side ----- \n");

    printf("Enter the received data words:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &rdata[i]);

        receivedSum = receivedSum + rdata[i];

        /* Add carry if generated */
        if(receivedSum > 255)
        {
            receivedSum = (receivedSum & 255) + 1;
        }
    }

    printf("Enter the received checksum: ");
    scanf("%d", &receivedChecksum);

    receivedSum = receivedSum + receivedChecksum;

    /* Add carry after adding checksum */
    if(receivedSum > 255)
    {
        receivedSum = (receivedSum & 255) + 1;
    }

    /* Verify checksum */

    if((unsigned char)(~receivedSum) == 0)
    {
        printf("\nNo Error Detected.");
        printf("\nData Accepted.");
    }
    else
    {
        printf("\nError Detected.");
        printf("\nData Rejected.");
    }

```

```
    }  
    getch();  
}
```

Output: